

SREERAM RADHAKRISHNAN

INDIAN INSTITUTE OF TECHNOLOGY MADRAS

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Education

Indian Institute of Technology, Madras <i>Bachelor of Technology in Electrical Engineering</i>	2023 – Present CGPA: 8.68/10.0
Chettinad Vidyashram <i>All India Senior School Certificate Examination Board</i>	2020 – 2022 XII: 97.2%
Atomic Energy Central School Anupuram <i>All India Secondary School Examination Board</i>	2008 – 2020 X: 96.2%

Publications

- Co-authored, “APEX: A Dual-Sparsity Accelerator for Precise and Efficient SNN Inference,” **ISVLSI 2026** [arXiv]. (Follow-up work under submission to HPCA 2027.)

Scholastic Achievements

- Gold Medalist** in the **VLSI Problem Statement** released by **ISRO**¹ at the 14th Inter IIT Tech Meet.
- Secured **All India Rank 890** in JEE Advanced 2023 (out of over 1.02 million JEE Main aspirants).
- Secured **All India Rank 747** and **99.9447 percentile** in JEE (Main) 2023.
- Secured a **100 percentile** in Physics in JEE Mains 2023 out of over 1.02 million candidates.

Internship Experience

Design of a Unified Multi-Precision Transcendental Function Unit for GPUs <i>Skills: SystemVerilog, Synopsys DC</i>	May 2026 – Jul 2026 <i>Agrani Labs, A stealth GPU startup</i>
- Designed the RTL of a shared, multi-precision (floating-point) transcendental function unit for GPUs, a unified block computing multiple transcendental functions across floating-point datatypes with shared hardware resources. - Took the design through the full flow, including microarchitecture, RTL, synthesis, post-synthesis netlist verification, and formal Datapath Verification (DPV) using Synopsys HECTOR, delivering a fully verified, synthesizable block optimized for power, performance, and area through hardware resource sharing across functions and precision formats.	

Relevant Coursework and Skills

Coursework : Embedded Memory Design, Computer Organization, Solid State Devices, Digital IC Design, CAD for VLSI, Secure Processor Microarchitecture, Machine Learning Foundations

Skills : SystemVerilog, Cadence Xcelium, Genus, Innovus, Calibre, Virtuoso, Tempus, LTSpice, Vivado HLS, AVR and ARM Assembly, C, C++, Python, Bash

Projects and Experiences

Design of a Digital Dual-Sparse Neuromorphic Accelerator <i>Skills: SystemVerilog, Synopsys DC, Python</i>	Jan 2026 – Present <i>Brainseek Lab, IIT Madras</i>
- Working on the architectural and microarchitectural design of a digital dual-sparse neuromorphic accelerator being built end-to-end with its accompanying software stack, under the guidance of Prof. Gopalakrishnan Srinivasan at Brainseek Lab, IIT Madras. - Designed a custom dimension-agnostic instruction set architecture (ISA) from scratch, tailored to the accelerator’s sparsity and precision requirements; mapped quantized SNN model weights and parameters to a unified, fully integer-arithmetic mixed-precision datapath, building the corresponding hardware to support quantized inference. - Optimized the processing element to achieve 3.17× higher energy efficiency than the state-of-the-art dual-sparse SNN accelerator without compromising latency or throughput. Co-authored as “APEX: A Dual-Sparsity Accelerator for Precise and Efficient SNN Inference” at ISVLSI 2026; additional innovations are under submission to HPCA 2027.	

¹Indian Space Research Organisation

- Ran a systematic **dataflow design-space exploration** for the accelerator's PE array, deriving a closed-form model of **on-chip SRAM traffic**, **off-chip traffic**, **storage**, and **energy costs** to identify an optimal schedule and array shape that substantially reduces SRAM reads at lower per-PE storage cost (validated across VGG-16, ResNet-18, and AlexNet); exploring **compiler-level techniques** that use this model to dynamically select the optimal dataflow per layer as workload demands change, rather than fixing a single dataflow for the entire network.

Inter IIT Tech Meet – ASIC Physical Design

Nov 2025 – Dec 2025

Skills learnt: Verilog, Xcelium, Genus, Innovus, Calibre, Virtuoso, Tempus

IIT Madras

- Implemented the **RTL-to-GDSII ASIC flow** using Cadence tools for synthesis, place-and-route, and post-layout simulations, achieving a **DRC- and LVS-clean** layout for the **decimation filter of a $\Delta\Sigma$ ADC** in **SCL180**.
- Performed **post-layout STA signoff** in Cadence **Tempus** across SS/FF PVT corners with single-mode and OCV analysis, achieving positive setup and hold slack across all modes.
- Optimized area and power through architectural and physical-design choices, achieving improved PPA with validated post-layout performance using back-annotated simulations.

Digital IC Design Lab

Jul 2025 – Nov 2025

Skills learnt: SystemVerilog, NGSpice, C, Python

IIT Madras

- Designed and simulated CMOS logic circuits (16-bit ripple carry adder, ring oscillator) in **NGSpice** with sky130 PDK.
- Created physical layouts in **KLayout**, performed **DRC** and **LVS** checks, and validated schematic-to-layout consistency.
- Implementation of **carry-look ahead adder architecture (Brent-Kung, Kogge-stone)** for delay and frequency.
- Implementation of a **16-bit carry-bypass adder in SystemVerilog and STA** to identify critical path and slack through constraint-based timing tools.

Enabling Use-Case Adaptation in Speech Recognition Systems

Sep 2024 – Sep 2025

Skills learnt: Python, Pytorch, Pandas, Bash

AI4Bharat Lab, IIT Madras

- Working under the guidance of **Prof. Mitesh Khapra** on developing an agentic framework for synthetic text generation to improve the adaptability of **Automatic Speech Recognition (ASR) systems** to specialized domains and **low-resource language** scenarios.
- Comparing a novel synthetic text generation technique against text only methods like Shallow fusion with external LMs, hotword boosting, and LLM rescoring to assess ASR performance improvements.

Project NightVision

May 2024 – Dec 2024

Skills learnt: Deep Learning, PyTorch, Numpy, Pandas

IIT Madras

- Worked on Low light image enhancement leveraging Computer Vision and various Deep Learning methods.
- Performed an extensive literature review on Retinex theory, Attention mechanism and various other methods.
- Implemented fundamental architectures like Vision Transformer, U-Net and GANs from scratch using PyTorch.
- Implemented a Retinex-based deep learning model featuring iterative illumination refinement and reflectance denoising for low-light image enhancement.

Leadership / Extracurricular Activities

AI Club Coordinator

May 2024 – Mar 2025

Skills learnt: Deep Learning, PyTorch, Numpy, Pandas

IIT Madras

- Conducted sessions on basics of Machine Learning and Computer Vision to 1000+ students from all over India.
- Conducted workshop at various colleges on Deep Learning and Advanced Vision architectures.
- Spearheaded a project on Image Captioning models exploring various architectures and implementing them.

Avanti Mentor

Nov 2023 – Oct 2024

Avanti Fellows

IIT Madras Chapter

- Mentored students from Jawahar Navodaya Vidyalaya, Puducherry and Barwani for JEE Mains and Advanced.
- Prepared mock examination papers to meet the specific requirements of diverse competitive examinations.

IViL Volunteer

Dec 2024 – Jan 2025

IIT for Villages

Shaastra & Saarang, IIT Madras

- Empowered rural women with business skills by managing the logistics and finances at the IViL (IIT for Villages) food stalls during the Shaastra and Saarang fests at IIT Madras.